1. (a) Evaluate the indefinite integral

$$\int \frac{e^{\sqrt{x}}\cos(\sqrt{x})}{\sqrt{x}}\,dx.$$

[7 marks]

(b) Using your answer from part (a), show that

$$\int_0^{\pi^2} \frac{e^{\sqrt{x}}\cos(\sqrt{x})}{\sqrt{x}}\,dx = -(1+e^{\pi}).$$

What is the geometric interpretation of the value of the integral in part (b)?

[3 marks]

[Total 10 marks]

Your Solution:

a) $\int \frac{e^{\sqrt{x}}\cos(\sqrt{x})}{\sqrt{x}}\,dx$

$u = \sqrt{x} = x^{\frac{1}{2}}$

$du = \frac{1}{2}x^{\frac{1}{2}-\frac{2}{2}} = \frac{1}{2}x^{-\frac{1}{2}} = \frac{1}{2}\frac{1}{\sqrt{x}} = \frac{1}{2\sqrt{x}}\,dx$

$(2\sqrt{x})\,du = \frac{1}{\cancel{2\sqrt{x}}}\,dx\,\cancel{(2\sqrt{x})}$

$\int \frac{e^{u}\cos(u)}{u}\,2u\,du$

$dx = du(2\sqrt{x})$

$dx = du(2u)$

$dx = 2u\,du$

$2\int \frac{e^{u}\cos(u)\,\cancel{u}}{\cancel{u}}\,du = 2\int e^{u}\cos(u)$

$2\left[e^{u}\sin(u) - \int e^{u}\sin(u)\,du\right]$

$u = e^{u}$

$du = e^{u}$

$v = \sin(u)$

$dv = \cos(u)$

$2e^{u}\sin(u) - \int e^{u}\sin(u)\,du$

$2e^{u}\sin(u) - (-e^{u}\cos(u)) - \int -e^{u}\cos(u)\,du$

$u = e^{u}$

$du = e^{u}$

$v = -\cos(u)$

$dv = \sin(u)$

Continue your solution to question 1 here…

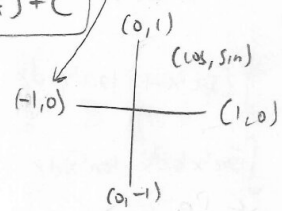
$$2\int e^u\cos(u) = 2e^u\sin(u) + e^u\cos(u) + \int e^u\cos(u)\,du$$

$$2\int e^u\cos(u) - \int e^u\cos(u) = 2e^u\sin(u) + e^u\cos(u)$$

$$\int e^u\cos(u) = 2e^u\sin(u) + e^u\cos(u) + C$$

$$\boxed{= \cancel{2}e^{\sqrt{x}}\sin(\sqrt{x}) + e^{\sqrt{x}}\cos(\sqrt{x}) + C}$$

$\frac{\pi}{1} \times \frac{180}{\pi} = 180^\circ$



$\cos(\pi) = -1$
$\sin(\pi) = 0$

$\cos(0) = 1$
$\sin(0) = 0$

b) $2e^{\sqrt{x}}\sin(\sqrt{x}) + e^{\sqrt{x}}\cos(\sqrt{x})\Big|_0^{\pi^2}$

$\sin\left(\sqrt{\frac{\pi}{2}}\right)$

$$\left[2e^{\pi}\cancel{\sin(\pi)} + e^{\pi}\cos(\pi)\right] - \left[2e^{\sqrt{0}}\sin(0) + e^{\sqrt{0}}\cos(0)\right]$$

$$e^{\pi}(-1) - \left[2e^{0}\cancel{\sin(0)} + e^{0}\cos(0)\right]$$

$$-e^{\pi} - [1]$$

$$-e^{\pi} - 1$$

$$\boxed{-(1+e^{\pi})}$$

Therefore the relationship hold.

c) The value of the definite integral in part b is the area under the curve of $\int \frac{e^{\sqrt{x}}\cos(\sqrt{x})}{\sqrt{x}}\,dx$ between π^2 and 0.

Q2 10

$\int \tan^2 x \cdot \tan^2 x \, dx \quad u =$

2. Evaluate the following integrals.

(a) $\int \tan^4 x \, dx.$ $\int \tan^2 x \cdot \tan^2 x$

[7 marks]

$\int$

(b) $\int_0^{\pi/4} \tan^4 x \, dx.$ $\int \tan^2 x \cdot [\sec^2 x - 1]$

[3 marks]

$u = \sec^2 x$
du

[Total 10 marks]

Your Solution:

$\tan^2 x + 1 = \sec^2 x$

$\tan^2 x = \sec^2 x - 1$

$\frac{\sin^2 x}{\cos^2 x} + \frac{\cos^2 x}{\cos^2 x} = \frac{1}{\cos^2 x}$

$\tan^2 x + 1 = \sec^2 x$

$\int [\sec^2 x - 1] \tan^2 x \, dx$

$u = \tan x$
$du = \sec^2 x \, dx$

$u = \sec x$
$du = \sec x \tan x \, dx$

$\int [\sec^2 x \tan^2 x - \tan^2 x] dx$

$dx = \frac{du}{\sec^2 x}$

$\int \sec^2 x \tan^2 x \, dx - \int \tan^2 x \, dx =$

$u = \tan x$
$du = \sec^2 x \, dx$
$dx = \frac{du}{\sec^2 x}$

$\int \sec^2 x \tan^2 x \, dx - \int \tan^2 x \, dx$

reversed.

$\frac{d}{dx} \tan x = \sec^2 x$

$\int \cancel{\sec^2 x} \cdot u^2 \frac{du}{\cancel{\sec^2 x}}$

$\int \tan^2 x \, dx$

$\int u^2 du$

$\int [\sec^2 x - 1] dx$

$\frac{u^3}{3} + c = \boxed{\frac{\tan^3 x}{3}} + c$

$\int \sec^2 x \, dx - \int 1 \, dx$

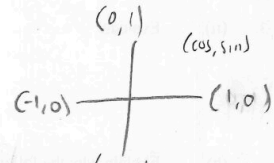
$\boxed{\tan x - x}$

Now: $\frac{\tan^3 x}{3} - [\tan x - x] + C$

$= \frac{\tan^3 x}{3} - \tan x + x + C$

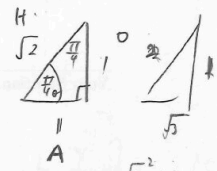
Continue your solution to question 2 here…

$$\frac{\tan^3 x}{3} - \tan x + x \Big|_0^{\pi/4}$$



$$\left[\frac{1}{3}\tan^3\left(\frac{\pi}{4}\right) - \tan\left(\frac{\pi}{4}\right) + \frac{\pi}{4}\right] - \left[\cancel{\frac{1}{3}\tan^3(0) - \tan(0) + 0}\right]$$

$$\frac{\sin(0)}{\cos(0)} = \frac{0}{1} = 0$$



$$\left[\frac{1}{3}(1)^3 - 1 + \frac{\pi}{4}\right]$$

$$\frac{1}{3} - \frac{1(12)}{1(12)} + \frac{\pi}{4}$$

$$\frac{4}{12} - \frac{12}{12} + \frac{3\pi}{12}$$

$$= \frac{-8}{12} + \frac{3\pi}{12}$$

$$\boxed{= \frac{3\pi - 8}{12}}$$

$1^2 + 1^2 = \sqrt{2}^2$

$2 = 2$ ✓

$\sqrt{3}^2 + 2$

$1^2 + \sqrt{3}^2 = 2^2$

$1 + 3 = 4$

$$\tan\left(\frac{\pi}{4}\right) = \frac{\sin\left(\frac{\pi}{4}\right)}{\cos\left(\frac{\pi}{4}\right)}$$

$$= \frac{\frac{1}{\sqrt{2}}\frac{\sqrt{2}}{\sqrt{2}}}{\frac{1}{\sqrt{2}}\frac{\sqrt{2}}{\sqrt{2}}} = 1$$

$$= \frac{\frac{\sqrt{2}}{2}}{\frac{\sqrt{2}}{2}} = 1$$

Q3 5

3. (a) Evaluate

$$\int \frac{x}{(x-3)^3}\,dx.$$

[4 marks]

(b) Explain why the following definite integral is an improper integral.

$$I = \int_1^3 \frac{x}{(x-3)^3}\,dx.$$

[2 marks]

(c) Show whether I is convergent or divergent. [4 marks]

[Total 10 marks]

Your Solution:

a) $\int \frac{x}{(x-3)^3}\,dx$ $\quad u = x-3$, $du = dx$

$u = x-3$
$+3 \quad +3$
$x = 3+u$

$$\int \frac{3+u}{u^3}\,du = \int \frac{3}{u^3}\,du + \int \frac{u}{u^3}\,du = 3\int \frac{1}{u^3}\,du + \int \frac{1}{u^2}\,du$$

$$= 3\int u^{-3}\,du + \int u^{-2}\,du$$

$$= 3\left[\frac{u^{-3+1}}{-3+1}\right] + \left[\frac{u^{-2+1}}{-2+1}\right] + C$$

$$= 3\left[\frac{u^{-2}}{-2}\right] + \left[\frac{u^{-1}}{-1}\right] + C$$

$$= \frac{3}{-2}u^{-2} + \left(-\frac{1}{u}\right) + C$$

$$= -\frac{3}{2u^2} - \frac{1}{u} + C$$

$$\boxed{-\frac{3}{2(x-3)^2} - \frac{1}{x-3} + C}$$

Continue your solution to question 3 here...

b) $\left[-\frac{3}{2(x-3)^2} - \frac{1}{x-3}\right]\Big|_1^3$

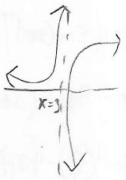
$\left[-\frac{3}{2(0)^2} - \frac{1}{3-3}\right] - \left[-\frac{3}{2(1-3)^2} - \frac{1}{1-3}\right] = \left[-\frac{3}{0} - \frac{1}{0}\right] + \left[\frac{3}{8} - \frac{1}{-2}\right]$

This definite integral is improper b/c when you plug in $x=3$, you get divide by zero, which is undefined

Need to look at integrand

c) The integral is divergent since it does not converge to a finite value. When you plug in the upper limit, it is undefined.

Show?



Rough sketch of rational function

Q4 8

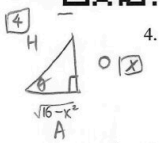
$\sin(2\theta) = 2\sin\theta\cos\theta$

$\sin^2\theta + \cos^2\theta = 1$

$\cos^2\theta = 1 - \sin^2\theta$

4. Consider the integral

$$I = \int \frac{x^2}{\sqrt{16 - x^2}}\,dx.$$



(a) What integration technique could be used to evaluate the integral I. [2 marks]

$x^2 + A^2 = 4^2$

$A^2 = 4^2 - x^2$

$a = \sqrt{16 - x^2}$

$\frac{x}{4} = \sin\theta$

Trig substitution, set $x = a\sin\theta$

(b) Evaluate the integral I. [6 marks]

[Total 8 marks]

$\theta = \sin^{-1}\left(\frac{x}{4}\right)$

Your Solution:

$a = \sqrt{16} = 4$

$\cos\theta = \frac{\sqrt{16 - x^2}}{4}$

$\sqrt{a^2 - x^2}$ $\quad x = a\sin\theta$

$\sqrt{x^2 - a^2}$ $\quad x = a\sec\theta$

$\sqrt{x^2 + a^2}$ $\quad x = a\tan\theta$

$$\int \frac{x^2}{\sqrt{16 - x^2}}\,dx$$

$x = 4\sin\theta,\ x^2 = 16\sin^2\theta$

$dx = 4\cos\theta\,d\theta$

$$\int \frac{16\sin^2\theta\, 4\cos\theta\,d\theta}{\sqrt{16 - 16\sin^2\theta}} = \int \frac{16\sin^2\theta\, 4\cos\theta\,d\theta}{\sqrt{16(1 - \sin^2\theta)}}$$

$$\int \frac{16\sin^2\theta\, 4\cos\theta\,d\theta}{\sqrt{16\cos^2\theta}} = \int \frac{16\sin^2\theta\, \cancel{4\cos\theta}\,d\theta}{\cancel{4\cos\theta}}$$

$\sin^2\theta = \frac{1}{2}\left[1 - \cos(2\theta)\right]$

$= \frac{1}{2} - \frac{1}{2}\cos(2\theta)$

$\int \cos(2\theta) = \frac{\sin(2\theta)}{2} + C$

$= \frac{1}{2}\sin(2\theta) + C$

$$16\int \sin^2\theta\,d\theta = 16\left[\frac{1}{2}\theta - \frac{1}{4}\sin(2\theta)\right]$$

$$16\int\left[\frac{1}{2} - \frac{1}{2}\cos(2\theta)\right]d\theta = 8\theta - 4\sin(2\theta) + C$$

$$16\left[\int\frac{1}{2}\,d\theta - \int\frac{1}{2}\cos(2\theta)\,d\theta\right] = 8\sin^{-1}\left(\frac{x}{4}\right) - 4\left[2\left(\frac{x}{4}\right)\left(\frac{\sqrt{16 - x^2}}{4}\right)\right]$$

$$16\left[\frac{1}{2}\theta - \frac{1}{2}\int\cos(2\theta)\,d\theta\right] = 8\sin^{-1}\left(\frac{x}{4}\right) - \cancel{4}\left[\frac{x}{2}\left(\frac{\sqrt{16 - x^2}}{\cancel{4}}\right)\right]$$

$$16\left[\frac{1}{2}\theta - \frac{1}{2}\left[\frac{1}{2}\sin(2\theta)\right]\right] \quad = 8\sin^{-1}\left(\frac{x}{4}\right) - \left[\frac{x}{2}\sqrt{16 - x^2}\right] + C$$

$\frac{d}{dx}\sin x = \cos x$

Q5 7

5. Evaluate the integral

$$\int \frac{2x+1}{x^2+2x-8}\,dx.$$

$1(-8) = -8$
$\neq -2$

[Total 7 marks]

Your Solution:

$$\int \frac{2x+1}{(x+4)(x-2)}\,dx = \int \frac{7/6}{(x+4)}\,dx + \int \frac{5/6}{(x-2)}\,dx = \frac{7}{6}\ln|x+4| + \frac{5}{6}\ln|x-2| + C$$

$$\cancel{(x+4)(x-2)}\left[\frac{(2x+1)}{\cancel{(x+4)(x-2)}} = \frac{A}{(x+4)} + \frac{B}{(x-2)}\right](x+4)(x-2)$$

$$2x+1 = A(x-2) + B(x+4)$$

$x=2$ $\quad 5 = A(2-2) + B(2+4)$

$$5 = B(6)$$

$$\boxed{B = 5/6}$$

$x=-4$ $\quad 2(-4)+1 = A(-4-2) + B(-4+4)$

$$-8+1 = A(-6) + \cancel{B(6)}$$

$$-7 = A(-6)$$

$$\boxed{A = \frac{-7}{-6} = \frac{7}{6}}$$

Continue your solution to question 5 here…

Q6 5

6. Find an implicit general solution of the following differential equation.

$$\frac{du}{dt} = \frac{1+t^4}{ut^2+u^4t^2}.$$

[Total 5 marks]

Your Solution:

$$(u+u^4)\quad \frac{du}{dt} = \frac{1+t^4}{t^2\cancel{(u+u^4)}}\cancel{(u+u^4)}$$

$$\cancel{dt}\frac{du}{\cancel{dt}}(u+u^4) = \frac{1+t^4}{t^2}dt$$

$$\int(u+u^4)du = \int\frac{1+t^4}{t^2}dt$$

$$\frac{u^2}{2}+\frac{u^5}{5}+c_1 = -\frac{1}{t}+\frac{1}{3}t^3+C_2$$

$$\frac{5u^2}{10}+\frac{2u^5}{10}+c_1 = -\frac{1}{t}+\frac{1}{3}t^3+C_2$$

$$\frac{5u^2+2u^5}{10}+c_1 = -\frac{1}{t}+\frac{1}{3}t^3+C_2$$

$$\frac{5u^2+2u^5}{10} = -\frac{1}{t}+\frac{1}{3}t^3+(C_2-C_1)$$

$$\frac{5u^2+2u^5}{10} = -\frac{1}{t}+\frac{1}{3}t^3+C$$

$$\int\frac{1+t^4}{t^2} = \int\frac{1}{t^2}dt+\int\frac{t^4}{t^2}dt$$

$$= \int\frac{1}{t^2}dt+\int t^2dt$$

$$= \int t^{-2}dt+\int t^2dt$$

$$= \frac{t^{-2+1}}{-2+1}+\frac{t^3}{3} = \frac{t^{-1}}{-1}+\frac{t^3}{3}$$

$$= \frac{1/t}{-1}+\frac{1}{3}t^3$$

$$= -\frac{1}{t}+\frac{1}{3}t^3$$

Q7 1

7. Consider the following differential equation,

$$x^2y' + 2xy = \ln x\,.$$

(a) Classify the differential equation as either *directly integrable differential equation*, *separable differential equation*, or *first-order linear differential equation*. Justify your answer. [2 marks]

(b) Solve the initial value problem,

$$x^2y' + 2xy = \ln x\,, \text{ where } y(1) = 2.$$

[8 marks]

[Total 10 marks]

Your Solution:

a) $x^2 \frac{dy}{dx} + 2xy = \ln x \Rightarrow \frac{dy}{dx} + \frac{2xy}{x^2} = \frac{\ln x}{x^2}$

$\frac{dy}{dx} + P(x)y = Q(x)$

$\frac{dy}{dx} + \frac{2y}{x} = \frac{\ln x}{x^2}$

$I(x) = e^{\int P(x)dx} = e^{\int 2\cdot\frac{1}{x}dx} = e^{2\int\frac{1}{x}dx}$

$= e^{2\ln|x|} = 2x$

$P(x) = \frac{2}{x}$

$Q(x) = x^{-2}\ln x$

$y = \frac{1}{I(x)}\left[\int I(x)Q(x)dx + c\right]$

$y = \frac{1}{2x}\left[\int 2x(x^{-2}\ln x)dx + c\right]$

$y = \frac{1}{2x}\left[\int 2x^{-1}\ln x\,dx + c\right]$

$y = \frac{1}{2x}\left[2\int x^{-1}\ln x\,dx + c\right]$

$\int \ln x = x\ln x - x + c$

$\int 1\cdot\ln x\,dx$

$u = \ln x$

$du = \frac{1}{x}dx$

$v = x$

$dv = dx$

$= x\ln x - \int 1\,dx$

$x\ln x - x + c$

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$\int u\,dv = uv - \int v\,du$

$\int x^{-1}\ln x\,dx$

$u = x^{-1}$
$du = -x^{-2}$
$v = x\ln x - x$
$dv = \ln x$

$x^{-1}\left[x^1\ln x - x^1\right] - \int -x^{-2}\left[x\ln x - x\right]dx$

$x^{-1+1}\ln x - x^{-1+1} + \int x^{-2}\left[x^1\ln x - x\right]dx$

$x^0\ln x - 1$

$\ln x - 1 + \int\left(x^{-2+1}\ln x - x^{-2+1}\right)dx = \ln x - 1 + \int x^{-1}\ln x - x^{-1}$

$= \ln x - 1 + \int x^{-1}\left[\ln x - 1\right]dx = \int \frac{1}{x}\left[\ln x - 1\right]dx$

$\cancel{\ln x + \int x^{-1}\ln x - x^{-1}\,dx}$

$\cancel{\ln x + x^{-1}\left[x\ln x - x\right] - \int -x^{-2}\left[x\ln x - x\right]dx}$

$\cancel{\ln x + \ln x - 1}$

First order ODE:

$\frac{dy}{dx} + P(x)y = Q(x)$

$y = \frac{1}{I(x)}\int I(x)Q(x)\,dx + C$

$x^2\frac{dy}{dx} + 2xy = \ln x$

$I(x) = e^{\int P(x)dx} = e^{2\int \frac{1}{x}dx} = e^{2\ln|x|} = 2x$

$\frac{dy}{dx} + \frac{2xy}{x^2} = \frac{\ln x}{x^2}$

$\frac{dy}{dx} + \frac{2y}{x} = \frac{\ln x}{x^2}$

$P(x) = \frac{2}{x}$

$Q(x) = \frac{\ln x}{x^2}$

$\cancel{y = \frac{1}{2x}\left[\int 2x\left(\frac{\ln x}{x^2}\right)dx + C\right]\left[\int \frac{\ln x}{x^2}dx + C\right]}$

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a) $y = \frac{1}{2x}\left[\int 2x \frac{\ln x}{x^2} dx + C\right]$

$y = \frac{1}{2x}\left[2\int \frac{x \ln x}{x^2} dx + C\right]$

$y = \frac{1}{x}\left[\int \frac{\ln x}{x} dx + C\right]$

$y = \frac{1}{x}\left[\int \underset{u}{\ln x} \cdot \underset{dv}{\frac{1}{x}} dx + C\right]$

Answer to Question A

$y = \frac{1}{x}\left[\frac{(\ln x)^2}{2} + C\right]$

$u = \ln x$

$du = \frac{1}{x} dx$

$v = \ln x$

$dv = \frac{1}{x} dx$

$\int \frac{1}{x} dx = \ln x$

$\int \frac{\ln x}{x} dx = (\ln x)^2 - \int \frac{\ln x}{x} dx$

$\int \frac{\ln x}{x} dx + \int \frac{\ln x}{x} dx = (\ln x)^2$

$2\int \frac{\ln x}{x} dx = (\ln x)^2$

$\int \frac{\ln x}{x} dx = \frac{(\ln x)^2}{2}$

b) $y(1) = 2$

$2 = \frac{1}{1}\left[\frac{\ln(1)^2}{2} + C\right]$

$2 = \left[\frac{1}{2}\ln(1)^2 + C\right]$

$2 = \left[\cancel{\frac{1}{2}(0)} + C\right]$

$\boxed{2 = C}$

$\ln(0) = \text{undefined}$

$\ln(1) = 0$

Solution to the IVP

$y = \frac{1}{x}\left[\frac{(\ln x)^2}{2} + 2\right]$